

AUTOSAR High-Level Design (HLD) Document

Section: 1.0 System Overview & Architecture

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Document: AUTOSAR_Powertrain_Control_HLD.pdf

Version: 1.0 (Draft)

Classification: Engineering Verification

1.0 System Architecture Overview

The Powertrain Domain Controller implements standard AUTOSAR 4.x Classic Platform architecture. The system consists of Application Software Components (SWC), Sensor Abstraction Managers, Actuator Interfaces, and Base Software (BSW) diagnostic services.

Core Architectural Entities:

- Component: EngineControlSWC (Core Engine Management)
- Component: SensorManager (Sensor Signal Processing)
- Component: ControlManager (Actuation & Strategy Coordinator)
- Component: DiagnosticManager (UDS & OBD Diagnostic Services)
- Component: Actuator (Physical Actuation Driver)

System Safety Level: ASIL B

Target Microcontroller: TC397 TriCore

Engine Control Software Component Specification

Section: 2.0 Engine Control SWC & Interfaces

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2.1 Component Specification: EngineControlSWC

The EngineControlSWC is responsible for calculating fuel injection quantity, spark timing, and managing torque demands from the accelerator pedal.

Component: EngineControlSWC

Interface: EngineDataInterface

P-Port: EngineDataPPort

R-Port: SensorDataRPort

Signals:

EngineSpeed, EngineTemperature, TargetTorque, ThrottlePosition

[Section A ? Provider Assignment]

EngineDataInterface provider = EngineControlSWC

Dependencies & Interactions:

Dependency: EngineControlSWC -> SensorManager

EngineControlSWC uses interface EngineDataInterface to publish internal state.

Runnable: Runnable_EngineSpeedCalc (Period: 10ms)

Event: TimingEvent_10ms

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3.1 Diagnostic Management Architecture

Diagnostic communication is managed by the DiagnosticManager component in coordination with BSW modules.

Components involved in diagnostic communication:

- DiagnosticManager (Diagnostic Application Handler)
- Dem (Diagnostic Event Manager)
- Dcm (Diagnostic Communication Manager)
- CanIf (CAN Interface Layer)

Interface: DiagnosticIf

P-Port: DiagResponsePPort

R-Port: DiagRequestRPort

Signals:

DtcStatus, DiagnosticSessionState, FaultRecordData

Functional Interaction:

DiagnosticManager calls Dcm for UDS diagnostic request processing and error routing.

DiagnosticManager -> Dem

System Functional Execution & Data Flows

Section: 4.0 End-to-End Functional Flows

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4.1 End-to-End Functional Flow: Powertrain Control Loop

The primary closed-loop functional flow across components executes as follows:

SensorManager -> EngineControlSWC -> ControlManager -> Actuator

Flow Description:

1. SensorManager reads crankshaft sensor and coolant sensor ADC values.
2. SensorManager -> EngineControlSWC delivers filtered sensor frames.
3. EngineControlSWC calculates torque request and actuator commands.
4. EngineControlSWC -> ControlManager coordinates limits and safety checks.
5. ControlManager -> Actuator triggers the injector drivers.

4.2 Auxiliary Communication Routing:

CommunicationManager sends data to EngineControlSWC.

Note: CommunicationManager provides gateway routing from external telemetry.

Sensor Manager Component & Interface Binding

Section: 5.0 Sensor Management & Interfaces

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5.1 Component Specification: SensorManager

The SensorManager aggregates analog sensor inputs, performs noise filtering, and exposes standardized physical values to consumer software components.

Component: SensorManager

Interface: SensorDataIf

P-Port: SensorDataPPort

Signals:

EngineSpeed, EngineTemperature, AmbientPressure, OilPressure

[Section B ? Conflicting Provider Assignment]

EngineDataInterface provider = SensorManager

Dependencies:

SensorManager provides interface SensorDataIf

SensorManager -> BswM